

WR-I Preprint Draft v0.2

Response-Defined World Spaces:
Finite-Resolution Equivalence and Identifiability of Global Realizations

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Abstract

We formulate a response-based framework for studying admissible global realizations that are indistinguishable under a declared family of observation protocols. A world-response structure consists of a class of admissible global realizations, a family of admissible protocols, and the corresponding response maps. Exact response equality induces a quotient world space. Identifiable quantities are precisely those that factor through this quotient, protocol refinement induces canonical quotient maps, and a gauge-completeness criterion separates representational redundancy from residual response non-identifiability. For metric gauge quotients we prove a localized finite-resolution separation result: on any compact admissible domain, joint point separation by a continuous protocol family implies finite protocol separation with a uniform positive response margin at every fixed nonzero resolution. We also show why ordinary finite-tolerance indistinguishability is not generally an equivalence relation. Examples illustrate finite refinement, gauge reduction, infinite-dimensional response fibres, and the dependence of identifiability on admissibility assumptions. The framework is ontologically neutral. It does not identify response equivalence with physical identity, and it makes no claim that observation creates, selects, or modifies a world.

1 Introduction

Inverse problems rarely begin with an unconstrained question of what reality “is.” They begin with a declared model class, a declared mode of access, and a map from candidate structures to observable data. When that map is not injective, the data determine an equivalence class rather than a unique hidden object. This elementary fact appears in mature theories of statistical and structural identifiability, in inverse boundary problems, and in results on observationally indistinguishable spacetimes [1, 2, 3, 4, 5, 8].

WR-I asks for a framework in which the compared objects are not merely parameters or coefficients but *admissible global realizations*. The word “world” is used only as shorthand for an element of such a declared class. The basic question is:

Given a class of admissible global realizations and a family of physically admissible protocols, which distinctions between realizations are actually resolved by their responses?

The answer is encoded by a response quotient. The quotient itself is standard mathematics; the purpose of WR-I is to make it the controlled interface between three layers that are often conflated: exact response identifiability, gauge redundancy, and finite-resolution distinguishability.

Methodological precursor: from BC-XI to WREH

The response-first viewpoint used here has a direct methodological precursor in *Boundary Compensation XI: The Inverse Isotypic Gap Problem and Finite-Resolution Response Equivalence Classes* (BC-XI) [7]. In BC-XI, a finite-resolution inverse response map could be non-injective on hidden operator structures, so the inverse object was naturally a response-equivalence class rather than a uniquely reconstructed hidden representative.

WR-I retains that methodological lesson but changes the mathematical carrier. The objects being compared are no longer hidden operators or finite-dimensional realizations inside Boundary Compensation. They are admissible global realizations $W \in \mathcal{W}$, compared through a declared family of admissible protocols. Schematically,

$$\begin{array}{ccc} \text{hidden structures} & \xrightarrow{\text{BC-XI}} & \text{finite-resolution response classes,} \\ \text{response classes} & \xrightarrow{\text{WR-I}} & \text{classes of admissible global realizations.} \end{array}$$

This is a bridge, not an identification of programmes. No BC-XI theorem is silently promoted into a statement about global worlds, spacetime, cosmology, or ontology.

The distinction can be summarized in two questions:

- BC-XI asks what a finite-resolution response determines about a hidden object.
- WR-I asks what a family of admissible responses determines about a possible global realization.

Contributions and claim ceiling

The paper makes five controlled contributions.

- (i) It defines a world-response structure $(\mathcal{W}, \mathcal{P}, \{R_P\})$ and the exact response quotient generated by a protocol bundle.
- (ii) It characterizes identifiable quantities by factorization through that quotient and organizes protocol refinement by canonical surjections between quotients.
- (iii) It separates gauge redundancy from residual response non-identifiability and gives an exact gauge-completeness criterion.
- (iv) It proves a localized finite-resolution separation proposition on compact admissible domains, avoiding any global compactness assumption on the entire physical state space.
- (v) It separates exact response equivalence from finite-tolerance indistinguishability, which is generally non-transitive.

The elementary factorization statements are not presented as new mathematical theorems. The finite-resolution proposition is a compactness consequence proved here for the WREH architecture; no historical priority claim is made for that abstract topological statement. The proposed novelty of WR-I is architectural and programme-level, not a claim to have invented observational equivalence, quotient spaces, structural identifiability, or gauge ambiguity.

WR-I does *not* claim that response-equivalent realizations are ontologically identical, that underlying reality does not exist, that measurement creates a world, that experiment rewrites the past, that the Universe branches, or that a response quotient is a spacetime or cosmological object.

2 Relation to prior work

2.1 Structural identifiability

Rothenberg’s classical formulation of identification studies whether distinct underlying structures induce the same observable probability law [1]. In dynamical systems, structural identifiability asks whether unknown parameters are uniquely recoverable from ideal input-output data. Differential-algebraic approaches, global identifiability tests, and identifiable reparametrizations form a mature literature [2, 3, 4, 5]. WR-I therefore makes no novelty claim for input-output indistinguishability, recoverable parameter combinations, or reparametrization as such.

A particularly close recent neighbour is Zhang’s *Mathematical Identifiability Geometry*, which explicitly treats structures with identical observation data as equivalence classes and studies geometric invariants of the resulting space [6]. Consequently, WR-I does not claim that the geometrization of observation-equivalence classes is new. Its narrower target is the package consisting of an admissible global-realization class, an explicit admissible protocol family, gauge separation, refinement maps, and a handoff to a distinct global-realizability problem.

2.2 Gauge ambiguity in inverse problems

Inverse boundary problems provide concrete examples in which the observation map is invariant under a known transformation group. In the anisotropic Calderón problem, boundary measurements determine the interior geometry only up to boundary-fixing changes of coordinates in the appropriate setting [11]. This is a domain-specific benchmark for the difference between representational redundancy and additional response non-identifiability.

2.3 Global spacetime underdetermination

Manchak showed that global spacetime structure in classical general relativity can remain observationally underdetermined in a robust sense [8]. Cinti and Fano later argued that the physical reasonableness of candidate indistinguishable spacetimes matters to the force of such conclusions [9]. For WR-I this is not a dispute to be resolved but a structural lesson: the admissible realization class must be part of the mathematical input. One cannot infer a physically meaningful non-identifiability result from an unrestricted class of formally constructible objects.

2.4 Global-section obstructions

Abramsky and Brandenburger formulate contextuality and non-locality through obstructions to global sections of compatible empirical data [10]. Their work is a major predecessor for the future WR-II question of whether locally or finitely compatible information admits a global realization. WR-I does not identify a global section of a measurement scenario with a global world realization; that bridge, if useful, must be formulated explicitly.

3 World-response structures

Definition 3.1 (World-response structure). *A world-response structure is a triple*

$$\mathbf{W} = (\mathcal{W}, \mathcal{P}, \{R_P\}_{P \in \mathcal{P}}),$$

where $\mathcal{W}$ is a declared class of admissible global realizations, $\mathcal{P}$ is a family of admissible observation protocols, and each protocol P has a response map

$$R_P : \mathcal{W} \rightarrow Y_P$$

into a protocol-dependent response space Y_P .

Remark 3.2 (Admissibility is part of the model). *The set $\mathcal{W}$ is not the class of all mathematically describable objects, and $\mathcal{P}$ is not the class of all mathematically definable readouts. Both are inputs whose physical or modelling justification must be supplied by an application.*

Definition 3.3 (Protocol bundle and joint response). *For $\mathcal{A} \subseteq \mathcal{P}$, define*

$$Y_{\mathcal{A}} := \prod_{P \in \mathcal{A}} Y_P, \quad R_{\mathcal{A}}(W) := (R_P(W))_{P \in \mathcal{A}}.$$

For the empty bundle, $Y_{\emptyset}$ is the one-point set.

Definition 3.4 (Exact response equivalence). *For a protocol bundle $\mathcal{A}$, define*

$$W_1 \sim_{\mathcal{A}} W_2 \iff R_{\mathcal{A}}(W_1) = R_{\mathcal{A}}(W_2).$$

Definition 3.5 (Response-defined world space). *The response-defined world space for $\mathcal{A}$ is*

$$\mathcal{Q}_{\mathcal{A}} := \mathcal{W} / \sim_{\mathcal{A}},$$

with quotient map

$$q_{\mathcal{A}} : \mathcal{W} \rightarrow \mathcal{Q}_{\mathcal{A}}, \quad q_{\mathcal{A}}(W) = [W]_{\mathcal{A}}.$$

4 Exact response quotient and identifiable quantities

Proposition 4.1 (Response quotient representation). *The relation $\sim_{\mathcal{A}}$ is an equivalence relation, and there is a unique bijection*

$$\overline{R}_{\mathcal{A}} : \mathcal{Q}_{\mathcal{A}} \longrightarrow \text{Im } R_{\mathcal{A}}$$

such that

$$R_{\mathcal{A}} = \overline{R}_{\mathcal{A}} \circ q_{\mathcal{A}}.$$

Proof. The relation is the kernel equivalence relation of $R_{\mathcal{A}}$. Define $\overline{R}_{\mathcal{A}}([W]_{\mathcal{A}}) = R_{\mathcal{A}}(W)$. Equality of classes is exactly what makes this definition well posed and injective; surjectivity onto the image and uniqueness follow from the definitions. $\square$

Remark 4.2. *As a bare set, $\mathcal{Q}_{\mathcal{A}}$ contains exactly the distinctions present in the joint response image. A topology, metric, smooth structure, measure, stratification, or physical interpretation on $\mathcal{Q}_{\mathcal{A}}$ requires additional assumptions.*

Definition 4.3 (Identifiable quantity). *A map $F : \mathcal{W} \rightarrow Z$ is $\mathcal{A}$ -identifiable if*

$$W_1 \sim_{\mathcal{A}} W_2 \implies F(W_1) = F(W_2).$$

Proposition 4.4 (Identifiability factorization). *A map $F : \mathcal{W} \rightarrow Z$ is $\mathcal{A}$ -identifiable if and only if there exists a unique map*

$$\overline{F} : \mathcal{Q}_{\mathcal{A}} \rightarrow Z$$

such that

$$F = \overline{F} \circ q_{\mathcal{A}}.$$

Proof. An $\mathcal{A}$ -identifiable map is constant on the fibres of $q_{\mathcal{A}}$, so $\overline{F}([W]_{\mathcal{A}}) = F(W)$ is well defined. The converse is immediate from the factorization, and uniqueness follows from surjectivity of $q_{\mathcal{A}}$. $\square$

5 Protocol refinement

Definition 5.1 (Refinement order). *For protocol bundles $\mathcal{A}, \mathcal{B} \subseteq \mathcal{P}$, write*

$$\mathcal{A} \preceq \mathcal{B} \iff \mathcal{A} \subseteq \mathcal{B}.$$

Proposition 5.2 (Refinement monotonicity). *If $\mathcal{A} \subseteq \mathcal{B}$, then*

$$[W]_{\mathcal{B}} \subseteq [W]_{\mathcal{A}} \quad \text{for every } W \in \mathcal{W}.$$

Moreover, there is a unique surjection

$$\pi_{\mathcal{B}\mathcal{A}} : \mathcal{Q}_{\mathcal{B}} \twoheadrightarrow \mathcal{Q}_{\mathcal{A}}, \quad [W]_{\mathcal{B}} \mapsto [W]_{\mathcal{A}},$$

satisfying

$$q_{\mathcal{A}} = \pi_{\mathcal{B}\mathcal{A}} \circ q_{\mathcal{B}}.$$

For $\mathcal{A} \subseteq \mathcal{B} \subseteq \mathcal{C}$,

$$\pi_{\mathcal{C}\mathcal{A}} = \pi_{\mathcal{B}\mathcal{A}} \circ \pi_{\mathcal{C}\mathcal{B}}.$$

Proof. Equality of all responses in $\mathcal{B}$ implies equality of the subfamily indexed by $\mathcal{A}$. The stated quotient map is therefore well defined and surjective; the commuting and composition identities follow directly. $\square$

Remark 5.3. *The directed hierarchy of response quotients is the interface to WR-II. WR-I does not yet infer global existence from a refinement chain and does not assign physical time or dynamics to the refinement parameter.*

6 Gauge redundancy and residual non-identifiability

Definition 6.1 (Response-invariant gauge action). *Let a group G act on $\mathcal{W}$. The action is response-invariant for $\mathcal{A}$ if*

$$R_P(g \cdot W) = R_P(W)$$

for every $g \in G$, $W \in \mathcal{W}$, and $P \in \mathcal{A}$.

Proposition 6.2 (Gauge factorization). *If the G -action is response-invariant for $\mathcal{A}$, then*

$$G \cdot W \subseteq [W]_{\mathcal{A}}$$

for every $W \in \mathcal{W}$, and $R_{\mathcal{A}}$ factors uniquely through the orbit quotient:

$$\mathcal{W} \xrightarrow{\pi_G} \mathcal{W}/G \xrightarrow{\tilde{R}_{\mathcal{A}}} Y_{\mathcal{A}}.$$

Proof. Response invariance gives $R_{\mathcal{A}}(g \cdot W) = R_{\mathcal{A}}(W)$, so each orbit lies inside one response class. Thus $\tilde{R}_{\mathcal{A}}(G \cdot W) := R_{\mathcal{A}}(W)$ is well defined and gives the required factorization. $\square$

Proposition 6.3 (Gauge-completeness criterion). *Assume the G -action is response-invariant for $\mathcal{A}$. The following are equivalent:*

(a) $[W]_{\mathcal{A}} = G \cdot W$ for every $W \in \mathcal{W}$;

(b) the induced map

$$\tilde{R}_{\mathcal{A}} : \mathcal{W}/G \rightarrow Y_{\mathcal{A}}$$

is injective.

Proof. If every response class is exactly one orbit, equal induced responses imply equal orbits. Conversely, Proposition 6.2 gives $G \cdot W \subseteq [W]_{\mathcal{A}}$; injectivity on $\mathcal{W}/G$ forces any $W' \in [W]_{\mathcal{A}}$ to belong to the same orbit as W . $\square$

Remark 6.4. When Proposition 6.3(b) fails, the observation system leaves distinct gauge orbits unresolved. *WR-I* calls this residual response non-identifiability; no ontological conclusion follows from that label alone.

7 Localized finite-resolution separation

Global compactness of $X = \mathcal{W}/G$ is unnecessarily strong for the finite-resolution question. Experimental design normally concerns a bounded or otherwise controlled admissible domain. We therefore localize the argument.

Assume that $X = \mathcal{W}/G$ is a metric space with metric d_X . Let $K \subseteq X$ be a compact admissible domain. Suppose each protocol induces a continuous map

$$\tilde{R}_P : X \rightarrow (Y_P, d_P),$$

and that the full admissible family separates points of K :

$$x \neq y, \quad x, y \in K \quad \implies \quad \exists P \in \mathcal{P} : \tilde{R}_P(x) \neq \tilde{R}_P(y).$$

Proposition 7.1 (Localized finite-resolution separation). *For every $\delta > 0$ there exists a finite protocol bundle*

$$\mathcal{A}_{K,\delta} = \{P_1, \dots, P_m\} \subseteq \mathcal{P}$$

and, whenever

$$K_\delta^{(2)} := \{(x, y) \in K \times K : d_X(x, y) \geq \delta\}$$

is nonempty, a constant $\eta_{K,\delta} > 0$ such that

$$x, y \in K, \quad d_X(x, y) \geq \delta$$

implies

$$\max_{1 \leq j \leq m} d_{P_j}(\tilde{R}_{P_j}(x), \tilde{R}_{P_j}(y)) \geq \eta_{K,\delta}.$$

If $K_\delta^{(2)}$ is empty, the separation statement is vacuous.

Proof. The set $K_\delta^{(2)}$ is closed in the compact space $K \times K$, hence compact. For every $P \in \mathcal{P}$ define

$$U_P = \{(x, y) \in K_\delta^{(2)} : d_P(\tilde{R}_P(x), \tilde{R}_P(y)) > 0\}.$$

Continuity makes U_P open in $K_\delta^{(2)}$. Point separation implies that the family $\{U_P\}_{P \in \mathcal{P}}$ covers $K_\delta^{(2)}$. Compactness therefore yields a finite subcover $U_{P_1}, \dots, U_{P_m}$.

Define

$$h(x, y) = \max_{1 \leq j \leq m} d_{P_j}(\tilde{R}_{P_j}(x), \tilde{R}_{P_j}(y)).$$

The function h is continuous and strictly positive on $K_\delta^{(2)}$. Hence it attains a positive minimum,

$$\eta_{K,\delta} := \min_{(x,y) \in K_\delta^{(2)}} h(x, y) > 0.$$

$\square$

Remark 7.2 (What compactness is actually needed). *The proof only requires compactness of the separated-pair set $K_\delta^{(2)}$, not compactness of the whole physical state space X . Local compactness of X alone does not produce the finite subcover needed for a global statement; one must still select a compact domain or otherwise prove compactness of the relevant pair set.*

Remark 7.3 (Resolution dependence). *The proposition is not a finite exact-reconstruction theorem. As $\delta \downarrow 0$, the required bundle size may grow and $\eta_{K,\delta}$ may tend to zero. The chosen metric, compact domain, continuity assumptions, and protocol admissibility are all application-level inputs.*

8 Finite tolerance is not an equivalence relation in general

Suppose each response space Y_P carries a metric d_P , and choose positive weights w_P for which

$$d_{\mathcal{A}}(W_1, W_2) = \sup_{P \in \mathcal{A}} w_P d_P(R_P(W_1), R_P(W_2))$$

is finite on the pairs under consideration. For $\varepsilon > 0$, define

$$W_1 \approx_{\mathcal{A}, \varepsilon} W_2 \iff d_{\mathcal{A}}(W_1, W_2) \leq \varepsilon.$$

Proposition 8.1 (Failure of transitivity). *The relation $\approx_{\mathcal{A}, \varepsilon}$ is generally not transitive and therefore does not generally define a quotient.*

Proof. For one scalar response choose

$$R(W_0) = 0, \quad R(W_1) = \frac{3}{4}\varepsilon, \quad R(W_2) = \frac{3}{2}\varepsilon.$$

Then $W_0 \approx_\varepsilon W_1$ and $W_1 \approx_\varepsilon W_2$, but $W_0 \not\approx_\varepsilon W_2$. □

Remark 8.2. *WR-I therefore reserves “equivalence class” for exact response equality unless an explicit equivalence closure or another justified coarse relation is introduced. Finite-precision objects are instead called tolerance fibres, response neighbourhoods, or uncertainty regions.*

9 Examples

Example 9.1 (Finite binary realization with gauge and hidden physical content). *Let*

$$\mathcal{W} = \{0, 1\}^4, \quad W = (a, b, g, h).$$

Let $G = \mathbb{Z}_2$ flip the label g and leave (a, b, h) fixed. Define

$$R_a(W) = a, \quad R_b(W) = b, \quad R_h(W) = a \oplus h.$$

With only R_a , there are two response classes of size eight. With (R_a, R_b) , there are four classes of size four and the physical variable h remains unresolved. With (R_a, R_b, R_h) , there are eight classes of size two, and each class is exactly one gauge orbit. Thus the induced response map is injective on $\mathcal{W}/G$.

An artificial readout $R_g(W) = g$ breaks gauge invariance and would make representatives distinguishable. The example illustrates why admissible protocols must be part of the model definition rather than an arbitrary set of mathematical readouts.

Example 9.2 (Smooth refinement from genuine ambiguity to pure gauge). *Let*

$$\mathcal{W} = S_x^1 \times S_g^1,$$

with gauge group $G = S^1$ acting by

$$\alpha \cdot (x, g) = (x, g + \alpha).$$

Then $\mathcal{W}/G \cong S_x^1$. Consider

$$R_c(x, g) = \cos x, \quad R_s(x, g) = \sin x.$$

For $\mathcal{A} = \{R_c\}$, the induced map $x \mapsto \cos x$ is non-injective, so generic response fibres contain two distinct gauge orbits. For $\mathcal{B} = \{R_c, R_s\}$, the induced map

$$x \mapsto (\cos x, \sin x)$$

is injective, and the remaining response classes are exactly the gauge orbits. Thus protocol refinement removes residual non-identifiability without removing gauge redundancy.

Example 9.3 (Continuous histories, dense observations, and admissibility). *Let*

$$\mathcal{W} = C([0, 1], \mathbb{R}),$$

and for each $t \in [0, 1]$ define

$$P_t(f) = f(t).$$

For a finite sampling set $\mathcal{A} = \{t_1, \dots, t_n\}$,

$$[f]_{\mathcal{A}} = \{g \in C([0, 1]) : g(t_i) = f(t_i), i = 1, \dots, n\} = f + V_{\mathcal{A}},$$

where

$$V_{\mathcal{A}} = \{u \in C([0, 1]) : u(t_i) = 0 \text{ for all } i\}.$$

The space $V_{\mathcal{A}}$ is infinite-dimensional, so every finite sampling bundle leaves an infinite-dimensional response fibre.

Let $D \subset [0, 1]$ be countable and dense. If two continuous histories agree on every point of D , continuity forces them to agree everywhere. Thus the countable protocol family $\{P_t : t \in D\}$ is jointly injective on this admissible realization class.

The conclusion fails after enlarging the admissible class. Choose $t_0 \in [0, 1] \setminus D$, let

$$f(t) = 0,$$

and define the bounded discontinuous history

$$g(t) = \begin{cases} 1, & t = t_0, \\ 0, & t \neq t_0. \end{cases}$$

Then f and g agree on every protocol in the dense family D but are not the same global realization. Thus the same protocol family can be identifying on one admissible class and non-identifying on a larger one.

10 Calderón problem as a gauge benchmark

The anisotropic Calderón problem supplies a concrete benchmark for Propositions 6.2–6.3. Boundary voltage-current data are invariant under the appropriate boundary-fixing coordinate transformations, and uniqueness results are therefore formulated up to that invariance [11]. In the language of WR-I, the known transformation orbit lies inside the response fibre; the substantive inverse problem is whether any additional ambiguity remains after passing to the quotient by that orbit.

This analogy is deliberately limited. WR-I does not claim that every gauge problem is a Calderón problem or that the abstract group G has a universal physical meaning. The example serves only to calibrate the distinction between representational redundancy and residual non-identifiability.

11 Prior-art and novelty ledger

Prior line	Established idea	WR-I boundary
Rothenberg (1971)	Identification and observational equivalence in parametric stochastic models.	No novelty claim for observational equivalence or identification as such.
Saccomani et al. (2003); Bellu et al. (2007); Villaverde et al. (2016)	Structural identifiability of dynamical models from ideal input-output data, including differential-algebraic and computational tests.	WR-I does not relabel standard parameter identifiability as world identifiability; it changes the carrier to an explicitly declared class of global realizations and an explicit protocol family.
Meshkat–Sullivant (2014)	Identifiable reparametrizations and parameter combinations using algebraic and graph-theoretic structure.	Factorization through identifiable combinations is prior art; WR-I uses a quotient language at the global-realization level.
Zhang (2026)	Geometric treatment of observation-equivalence classes, including indistinguishability strata and unique-identification points.	WR-I does not claim that geometry of observation classes is new.
BC-XI (2026)	Finite-resolution response equivalence as the inverse object when hidden operator structures are not uniquely reconstructed.	Methodological and historical precursor. WR-I changes the carrier from hidden operator structures to admissible global realizations; no physical or ontological BC claim is imported.
Manchak (2009)	Observational underdetermination of global spacetime structure in classical GR.	Domain-specific predecessor for global-world underdetermination; WR-I abstracts the observation mechanism into a declared protocol family.

Prior line	Established idea	WR-I boundary
Cinti–Fano (2021)	Critique stressing physical reasonableness of observationally indistinguishable spacetimes.	Supports treating $\mathcal{W}_{\text{admissible}}$ as part of the formal input.
Abramsky– Brandenburger (2011)	Contextuality/non-locality as obstruction to global sections of local empirical data.	Major predecessor for WR-II. WR-I does not equate global sections with global world realizations.
Anisotropic Calderón inverse problem	Boundary response invariance under boundary-fixing coordinate changes and uniqueness up to the corresponding gauge in suitable settings.	Benchmark for gauge factorization and the question whether response fibres are strictly larger than gauge orbits.

The defensible novelty claim at v0.2 is therefore architectural:

WR-I formulates a protocol-indexed framework whose primary carrier is a declared class of admissible global realizations, separates gauge redundancy from residual response non-identifiability, organizes protocol refinement by canonical quotient maps, and connects ideal joint separation to finite experimental separation on compact admissible domains at fixed nonzero resolution. This structure is designed to feed a distinct global-realizability problem rather than to assert a preferred ontology.

The claim remains subject to continuing literature audit. In particular, no priority claim is attached to the elementary quotient factorization statements or to the general use of compactness in finite subcover arguments.

12 Discussion: three levels that must remain distinct

The framework separates three questions.

Exact response identifiability. For a fixed protocol bundle, exact equality of responses produces a genuine equivalence relation and quotient. This is an idealized structural question.

Gauge-reduced identifiability. If a declared gauge group acts response-invariantly, one must first determine whether the observation system distinguishes gauge orbits. Failure of injectivity on $\mathcal{W}/G$ is stronger than mere representational redundancy.

Finite-resolution distinguishability. Real experiments have finite accuracy and operate on controlled domains. The localized finite-resolution result therefore concerns a compact admissible region K and a nonzero resolution scale δ , not a global exact reconstruction claim.

These distinctions are also a claim firewall. Neither persistent response multiplicity nor finite-resolution ambiguity implies that multiple worlds are physically instantiated. Conversely, a unique point of a declared quotient is uniqueness relative to the chosen realization class and protocol family; it is not automatically uniqueness of ontology.

13 Handoff to WR-II

WR-I ends where a new mathematical question begins. Suppose progressively richer finite protocol bundles produce compatible response constraints. Even if every finite stage is nonempty, one may ask whether there exists any admissible global realization satisfying all stages simultaneously.

Schematically, refinement gives

$$\mathcal{Q}_1 \longleftarrow \mathcal{Q}_2 \longleftarrow \mathcal{Q}_3 \longleftarrow \cdots .$$

WR-II will study the distinction between

finite compatibility and global realizability.

Potential tools include compactness principles, inverse/projective limits, extension theorems, explicit counterexamples, and comparison with sheaf-theoretic global-section obstructions. None of these is assumed by WR-I. In particular, protocol refinement is not physical time, and failure of global realizability is not interpreted as destruction or nonexistence of the physical Universe without a separate bridge.

Conclusion

WR-I provides a deliberately modest mathematical interface. Responses define which distinctions survive a declared observation system; gauge structure identifies a known class of redundant distinctions; compact-domain separation relates ideal point separation to finite experimental design; and admissibility controls what counts as a candidate global realization in the first place.

The intellectual route from BC-XI to WR-I is therefore continuous but not identical: BC-XI established a response-first inverse viewpoint for hidden finite-resolution structures, whereas WR-I lifts the question to classes of admissible global realizations. The next step is not a stronger ontological claim but the mathematical question of global completion.

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